# Principle of Statistical Mechanics

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# 1 Entropy

Quantum mechanics is describes the true nature of physical reality. It governs the motion of atoms, the nature of chemical bonds, and the interaction of matter with light at the microscopic scale. However, our practical interests often lie in the properties of bulk matter at the human scale. When we observe a material, we want to know its macroscopic traits: Is it hot? Is it volatile? Is it elastic? Why is it lustrous? Since solving the Schrödinger equation for $10^{23}$ particles is computationally impossible, we need a strategy to recover these properties without intractable calculation. This is the role of statistical mechanics. It explains how macroscopic phenomena arise from the statistical distribution of atoms, effectively bridging the gap between quantum mechanics and thermodynamics. Ultimately, it offers us a first-principles understanding of the material world around us.

If we have complete knowledge of a system, we can specify its state $|\psi\rangle$, which is a fundamental postulate of quantum mechanics that we have all accepted. However, in reality, we do not have complete knowledge of the system, and we incoprate our own ignorance by saying that the system could be in a collection of these call pure state or micro state $\{|\psi_i\rangle\}$ with a probability distribution $\{p_i\}$. We want to emphasis immediately that the probability due to our own ignorance is philosophically different from the inherent probabilistic nature of quantum state that are modulus squares of their quantum amplitude. Our goal is to deal with our own ignorance and find an expression for $\{p_i\}$.

The the expectation value of some observable is

$$\langle \hat{O} \rangle = \sum_i p_i \langle \psi_i | \hat{O} | \psi_i \rangle \tag{1.1}$$

where this is the expectation value of the an observable $\hat{O}$ in some quantum state $|\psi_i\rangle$ weighed by the probability distribution $p_i$ due to our own ignorance.

Equivalently

$$\langle \hat{O} \rangle = \sum_i p_i \sum_j |a_{ij}|^2 o_j \tag{1.2}$$

where

$$\hat{O}|o_j\rangle = o_j |o_j\rangle \tag{1.3}$$

$$|\psi_i\rangle = \sum_j a_{ij} |o_j\rangle \tag{1.4}$$

A more succinct way to express the same thing is

$$\langle \hat{O} \rangle = \mathrm{tr}\left[\hat{\rho}\hat{O}\right] \tag{1.5}$$

where

$$\hat{\rho} = \sum_i |\psi_i\rangle p_i \langle \psi_i| \tag{1.6}$$

is known as the **density operator**.

Let us begin with the simplest case: an isolated system at equilibrium, characterized by a fixed number of particles $N$, volume $V$, and energy $E$:

Here, isolated implies that the system exchanges neither energy nor matter with its surroundings. Equilibrium implies that the system's probability distribution $\{p_i\}$ is time-independent. The probability distribution associated with constant $N, V, E$ is known as the **microcanonical ensemble**.

Energy means the eigenvalue of the system's Hamiltonian.

$$\hat{H}|\psi_i\rangle = E_i|\psi_i\rangle \tag{1.7}$$

If the only thing we know about the system is its energy, then there is no reason why we should believe that the system is more likely to be in one specific state $|\psi_i\rangle$. We have arrived at our first postulate

***The postulate of equal a priori probability: For an isolated system in equilibrium, all accessible microstates are equally likely.***

For now, we just have an energy constraint, if the number of mircrostates with energy $E$ is $\Omega(E)$, then

$$p_i = \frac{1}{\Omega(E_i)} \tag{1.8}$$

and the probability of the system in some other state with different energy $E' \neq E$ is zero. Note that in this case $\Omega(E_i) = g_i$ the degeneracy of energy level $i$.

It is worth pointing out that $\Omega$ is a mind-bollgingly large number, say, for a box of gas molecules with 10 states each, then the number of microstate of the box is $10^{10^{23}}$, compared with the number of atoms in the universe, which is only $10^{80}$, and typically a molecule has much more than 10 accessible states.

To proceed we consider a simple coin flip experiment, which the number of coins that land on their head follows a binomial distribution $X \sim B(n, p)$ where $n$ is the number of coin flips and $p = 1/2$ is the probability to land on head. Notice that both the mean $\mu = np = n/2$ and variance $\sigma^2 = np(1-p) = n/4$ increases linearly with $n$, thus the width of the distribution $\sigma = \sqrt{np(1-p)} = \sqrt{n}/2$. Therefore the relatively width of the distribution scales as $\sigma/\mu \sim 1/\sqrt{N}$ which approaches 0, as $N$ becomes large. Since $N \sim 10^{23}$ in our context, the distribution is a spike at the mean value $\mu$. We have considered a binomial distribution, but this holds for other types of distributions, and our system will only be found in what is known as the dominating configuration, and $N \to \infty$ is known as the **thermodynamics limit**.

we have arrived at our second postulate

***The observed macrostate of a system is the one containing the largest number of microstates***

We now define **entropy** as

$$S = k_B \ln \Omega \tag{1.9}$$

this is the simplest definition of entropy, and it is the one that is engraved on Boltzmann's gravestone. $k_\mathrm{B} = 1.381 \times 10^{-23}\,\mathrm{J \cdot K^{-1}}$ is the Boltzmann constant. We will see later, how is this definition of entropy connected to entropy in information theory and the thermodynamic definition of entropy in terms of heat transfer. We note that $S = S(E)$, it is a function of the energy of our system through the $\Omega(E)$ dependence. We have already established that $\Omega \sim e^N$, where $N$ is the number of particles, thus $S \sim N$ which means that entropy of different systems adds then we combine them, this is an important property.

Let's now consider two systems two isolated systems so that the total energy is $E = E_1 + E_2$, and

$$\Omega^0(E) = \Omega_1(E_1)\Omega_2(E_2) \tag{1.10}$$

$$S^0 = S_1 + S_2 \tag{1.11}$$

where we have noted the two systems with label 1 and 2.

If now put the two systems in contact with each other and allow them to exchange energy, we have

$$\Omega(E) = \sum_i \Omega_1(E_i)\Omega_2(E - E_i) \geq \Omega^0(E) \tag{1.12}$$

where we have noticed that there is no reason for the first system to stay in energy $E_1$ and second system to have energy $E_2$ once they are in contact with each other, but we must still hold the total energy constant. Eq. (1.7) now is just one particular term in the sum.

and, since logarithm is a monotonic function,

$$S \geq S^0 \tag{1.13}$$

This is the second law of thermodynamics in essence, it states that the system spread from an improbable configuration to the dominating configuration overtime, and it is not reversible in the macroscopic scale. This is what we mean by arrow of time.

Using the second postulate, we now find the domination configuration in which energy of the first system is $E^\star$, whilst keeping the total energy constant

$$\left(\frac{\partial(\Omega_1(E_1)\Omega_2(E - E_1))}{\partial E_1}\right)_E = 0 \tag{1.14}$$

after a bit of algebra, yields

$$\frac{1}{\Omega_1}\frac{\partial\Omega_1}{\partial E_1} = \frac{1}{\Omega_2}\frac{\partial\Omega_2}{\partial E_2} \tag{1.15}$$

or equivalently

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2} \tag{1.16}$$

the relation above holds, provided that our system is at equilibrium where $\Omega$ and $S$ of the combined system is maximised, and the system sits in the dominating configuration.

## 2 Temperature

We all have an intuitive feeling for temperature, when you have a fever, the mercury in the thermometer goes up when you it under your arm, you shiver when you didn't put on enough clothes when you go outside in winter, you quickly retrieve your hand when you touch a stove is on. In this chapter, we will talk about how temperature is related to the microscopic picture

**Temperature** is defined as

$$T = \Big(\frac{k_B \partial \ln \Omega}{\partial E}\Big)^{-1} \tag{2.1}$$

or equivalently

$$\beta = \frac{\partial \ln \Omega}{\partial E} \tag{2.2}$$

and our first goal is to show this is consistent with our intuition.

Directing substituting the definition into Eq. (I.13) yields, at equilibrium

$$T_1 = T_2 \tag{2.3}$$

which is what we expect and is the essence of the zeroth law of thermodynamics. Temperature a transitive property at equilibrium. Where transitive we mean if we have a third system in equilibrium with system 2, it will also be at equilibrium with system 1 with the same temperature.

We have fixed the energy of our system to $E$, let's relax this constraint and let the energy to fluctuation about its mean value and force the system to have temperature $T$. The probability distribution associated with constant $N, V, T$ is called the **canonical ensemble**.

and see how this changes the probability distribution $\{p_i\}$.

The way to keep our system at constant temperature is to immerse it into a massive heat bath, or reservoir that is set at the desired temperature $T$. Note that the total energy $E = E_{\text{sys}} + E_{\text{res}}$ and $E_{\text{res}} \gg E_{\text{sys}}$. Where the total energy is constant, but the energy of the system isn't.

Therefore we have,

$$p_i = p(E_{\text{sys}} = E_i) \propto \Omega_{\text{res}}(E - E_i) \tag{2.4}$$

where we have assumed energy eigenstates are non-degenerate.

Consider the following Taylor series expansion up to the linear term around $E$

$$\ln \Omega(E - E_i) = \ln \Omega(E) - E_i \frac{\partial \ln \Omega(E)}{\partial E} \tag{2.5}$$

using the definition of temperature Eq. (2.1) yields

$$\ln \Omega(E - E_i) = \ln \Omega(E) - \frac{E_i}{k_B T} \tag{2.6}$$

after a slight rearrangement, we have

$$\Omega(E - E_i) = \Omega(E) e^{-E_i/k_B T} \tag{2.7}$$

comparison with Eq. (2.3) yields

$$p_i = \frac{1}{Q} e^{-E_i/k_B T} = \frac{1}{Q} e^{-\beta E_i} \tag{2.8}$$

where $Z$ is a noramlising constant to ensure $\sum_i p_i = 1$, and $\beta = 1/k_B T$

If we repeat the argument from Eq. (2.3) but now assuming the energy levels are degenerate with degeneracy $g_i$, then we can generalise the result to

$$p_i = \frac{1}{Z} g_i e^{-E_i/k_B T} \tag{2.9}$$

A more elegant way to write this is

$$Q = \mathrm{tr}\left[e^{-\beta \hat{H}}\right] \tag{2.10}$$

$$\hat{\rho} = \frac{1}{Q} e^{-\beta \hat{H}} \tag{2.11}$$

## 3 Energy

Turns out, this harmless looking normalisation constant is the central object of statistical mechanics.

It is called the **partition function**, and the reason it plays such a central role is that it connects quantum mechanical energy levels to thermodynamic state functions.

$$Q = \sum_i e^{-E_i/k_B T} \tag{3.1}$$

Notice that we allowed allowed energy in our system to fluctuate about its mean value, which is how we define **internal energy**

$$U = \langle E \rangle = \sum_i p_i E_i \tag{3.2}$$

it is easy to show that

$$U = -\frac{\partial \ln Q}{\partial \beta} = k_B T^2 \frac{\partial \ln Q}{\partial T} \tag{3.3}$$

subtly, we have held the volume constant during differentiation, as energy eigenvalues depends on the volume in the sense of the boundary condition for the wavefunction to vanish.

The energy fluctuation is captured by the variance

$$\sigma_E^2 = \langle (E - \langle E \rangle)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 \tag{3.4}$$

it is easy to show that

$$\sigma_E^2 = \frac{\partial^2}{\partial \beta^2} \ln Q = -\frac{\partial U}{\partial \beta} = k_B T^2 C_V \tag{3.5}$$

where the isochoric heat capacity is defined as

$$C_V = \Big(\frac{\partial U}{\partial T}\Big)_V \tag{3.6}$$

Thus, how much heat you should provide to raise the temperature of a system is related to its energy fluctuation, which is not obvious.

We are now going to revise our definition of entropy. Notice that our original definition of entropy Eq. (1.6) varies with energy fluctuation $E$, we would like to find a new definition of entropy that does not fluctuate with energy.

Suppose we have a large number of copies $W$ of our system, then the number of systems with energy $E_i$ is $p_i W$.

The number of microstate is given by

$$\Omega = \frac{W!}{\prod_i (p_i W)!} \tag{3.7}$$

where the denominator removes the over counting, in the sense that switching around two copies of the same energy does not result in a new microstate.

The entire collection of the system has energy $WU$, where the fluctuation in energy vanishes provided that $W$ is large, therefore the total system has constant energy, and we can use our original definition of entropy.

Thus we have

$$S_{\text{tot}} = k_B \ln \Omega = k_B \ln \frac{W!}{\prod_i (p_i W)!} \tag{3.8}$$

Using Stirling's approximation

$$\ln W! = W \ln W - W \tag{3.9}$$

yields

$$S_{\text{tot}} = -W k_B \sum_i p_i \ln p_i \tag{3.10}$$

since the systems are not interacting, the entropy is purely additive thus for a single system we have

$$S = -k_B \sum_i p_i \ln p_i \tag{3.11}$$

which is the same entropy as in information theory known as the Shannon entropy $s$ (up to a multiplicative constant $k_B$ which carries the unit). This is in fact a more general definition of entropy, and it is obvious that the Boltzmann definition of entropy is a special case of this when you force the system to be at some energy $E_i$, so that $p_{j\neq i} = 0$

More elegantly

$$S = -k_B \operatorname{tr}[\hat{\rho} \ln \hat{\rho}] \tag{3.12}$$

It is then easy to show that

$$S = \frac{U}{T} + k_B \ln Q = k_B \frac{\partial}{\partial T}(T \ln Q) \tag{3.13}$$

notice the similarly between the second part on RHS and our original definition of entropy Eq. (1.6). Suggesting that the role played by the partition function $Z$ is the same as $\Omega$, the number of accessible microstates.

rearrangement the first part on RHS yields

$$-k_B T \ln Q = U - TS \tag{3.14}$$

we then define

$$A = -k_B T \ln Q \tag{3.15}$$

as the **Helmholtz free energy**, this is the thermodynamic state function that has the most intimate relation to the partition function, and it is through it, that we find other thermodynamic state functions. Its definition coincides with the definition in classical thermodynamics.

$$A = U - TS \tag{3.16}$$

The first term on RHS of Eq. (3.12) can be written as

$$\frac{1}{k_B} S = \beta U + \ln Q \tag{3.17}$$

taking the diffrential at constant volume yields

$$\frac{1}{k_B}dS = \beta dU + Ud\beta + \frac{\partial \ln Q}{\partial \beta}d\beta \tag{3.18}$$

using Eq. (3.3) yields

$$dS = \frac{1}{T}dU \tag{3.19}$$

thus we have following relation

$$T = \Big(\frac{\partial U}{\partial S}\Big)_V \tag{3.20}$$

which neatly related the three thermodynamic state functions: Temperature, internal energy and entropy, which are introduced by zeroth, first and second law of thermodynamics respectively.

Taking the differential of internal energy yields

$$dU = \sum_i \big(E_i dp_i + p_i dE_i\big) \tag{3.21}$$

Taking the differential of entropy yields

$$dS = -k_B \sum_i (\ln p_i + 1)dp_i = -k_B \sum_i \ln p_i dp_i \tag{3.22}$$

then using Eq. (2.7) we obtain

$$TdS = \sum_i E_i dp_i \tag{3.23}$$

At constant volume $dE_i = 0$, therefore, no work is done on the system. However, $dp_i \neq 0$ as $p_i$ depends on the temperature, thus only heat is transferred to the system.

More subtly, $p_i$ is obtained by maximising entropy for a given macrostate, thus it is the probability distribution for a system at equilibrium, therefore the heat is transferred reversibly and we identify

$$đq_{\text{rev}} = \sum_i E_i dp_i \tag{3.24}$$

$$dS = \frac{đq_{\text{rev}}}{T} \tag{3.25}$$

which is just the definition of entropy in classical thermodynamics.

We now define pressure

$$P = -\sum_i p_i \frac{dE_i}{dV} \tag{3.26}$$

multiply both sides by $dV$ yields

$$-PdV = \sum_i p_i dE_i \tag{3.27}$$

More subtly, given that $V$ is slowly varying so that the $\hat{H}$ of the system is slowly varying, a well known theorem called the 'Adiabatic theorem' from time-dependent perturbation theory states that the system will stay in whatever stationary state $|i\rangle$ it started in. Therefore, $p_i$ remains constant and $dp_i = 0$. This implies, no heat is transferred and entropy of the system does not change, this is precisely the meaning of reversible adiabatic process, hence the name of the theorem.

$$đw_{rev} = \sum_i p_i dE_i = -PdV \tag{3.28}$$

equivalently, pressure can be defined as

$$P = k_B T \frac{\partial \ln \Omega}{\partial V} = \frac{1}{\beta} \frac{\partial \ln \Omega}{\partial V} \tag{3.29}$$

we can then repeat the argument of Eq. (1.11) onward to obtain at equilibrium

$$P_1 = P_2 \tag{3.30}$$

this is known a the mechanical equilibrium.
Substituting Eq. (4.3) and (4.7) into (4.1) yields the fundamental equation

$$dU = TdS - PdV \tag{3.31}$$

thus we can obtain another expression for pressure

$$P = -\Big(\frac{\partial U}{\partial V}\Big)_S \tag{3.32}$$

which is the definition of pressure in classical thermodynamics. However, the most convenient form of pressure is from Helmholtz free energy,

$$dA = -SdT - PdV \tag{3.33}$$

then,

$$P = -\Big(\frac{\partial A}{\partial V}\Big)_T \tag{3.34}$$

At the end, we shall derive the ideal gas equation from the quantum energy levels of particles in a box, which is,

$$E_n = \frac{n^2 h^2}{8mL^2} \tag{3.35}$$

where $L$ is the length of 1D box,

$$\begin{aligned} Q &= \sum_{n=1}^{\infty} e^{-\beta E_n} \\ &= \sum_{n=1}^{\infty} e^{-\beta n^2 h^2/8mL^2} \\ &\simeq \int_1^{\infty} dn\, e^{-\beta n^2 h^2/8mL^2} \\ &\simeq \int_0^{\infty} dn\, e^{-\beta n^2 h^2/8mL^2} \\ &= \frac{1}{2}\int_{-\infty}^{\infty} dn\, e^{-\beta n^2 h^2/8mL^2} \\ &= \frac{1}{2}\sqrt{\frac{8\pi mL^2}{\beta h^2}} \\ &= \sqrt{\frac{2\pi m k_B T L^2}{h^2}} \\ &= \frac{L\sqrt{2\pi m k_B T}}{h} \end{aligned} \tag{3.36}$$

then define Broglie wavelength $\Lambda(T)$ as,

$$\Lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}} \tag{3.37}$$

therefore,

$$Q = \frac{L}{\Lambda(T)} \tag{3.38}$$

there is no interaction between 3 Carteisan directions, so in 3D,

$$Q = \frac{V}{\Lambda(T)^3} \tag{3.39}$$

where V is the volume of the box. For $N$ non-interacting and indistinguishable gas molecules, we have,

$$Q = \left(\frac{V}{\Lambda(T)^3}\right)^N \frac{1}{N!} \tag{3.40}$$

Clearly,

$$A = -k_B T \ln Q = -k_B T \left[\ln\left(\frac{V}{\Lambda(T)^3}\right)^N \frac{1}{N!}\right] \tag{3.41}$$

$$A = -k_B T \left[N \ln V - 3\ln\Lambda - 3\ln N!\right] \tag{3.42}$$

therefore,

$$P = -\left(\frac{\partial A}{\partial V}\right)_T = \frac{N k_B T}{V} \tag{3.43}$$

the number of mole is,

$$n \equiv \frac{N}{N_A} \tag{3.44}$$

and gas constant is,

$$R \equiv N_A k_B \tag{3.45}$$

thus,

$$PV = nRT \tag{3.46}$$

which is the ideal gas equation.

# 4 Chemical Potential

So far, we have discussed the relation between variation in energy and temperature, variation in volume and pressure in the previous chapter and now we will consider the variation in particle number $N$, or mol of substance $n = N/N_A$. This is extremely important, as interesting physical processes such as phase transition and chemical reaction, all involve the change in the amount of substance of different species.

We define chemical potential as

$$\mu = -k_B T \frac{\partial \ln \Omega}{\partial N} = -\frac{1}{\beta} \frac{\partial \ln \Omega}{\partial N} \tag{4.1}$$

where now both the volume and energy are held constant, repeating the argument of Eq. (1.11) again yields at equilibrium

$$\mu_1 = \mu_2 \tag{4.2}$$

which is known as chemical equilibrium, and there is by definition no net exchange of the number of particles between the two systems.

In the previous two ensembles, we have assumed both the particle number and volume are constant, i.e. the NVE/microcannonical ensemble in chapter one and NVT/canonical ensemble in chapter two onwards. Now we allow the number of particle of vary but fixing the chemical potential. The probability distribution at constant $\mu$, $P$, $T$ is called the **grand canonical ensemble**.

It follows that that the grand canonical partition function is

$$\mathcal{Z} = \sum_i e^{-\beta(E_i - \mu N_i)} \tag{4.3}$$

or more succinctly as

$$\mathcal{Z} = \mathrm{tr}\left[e^{-\beta(\hat{H} - \mu \hat{N})}\right] \tag{4.4}$$

to recover the previous equation, we recognise that in order to specify energy the particle number for a state $|E_i, N_i\rangle$, we must have $[\hat{H}, \hat{N}] = 0$.

The grand canonical potential plays the same role as Helmholz free energy in canonical ensamble

$$\Phi = -k_B T \ln \mathcal{Z} \tag{4.5}$$

and

$$p_i = \frac{1}{\mathcal{Z}} e^{-\beta(E_i - \mu N_i)} \tag{4.6}$$

$$\hat{\rho} = \frac{1}{\mathcal{Z}} e^{-\beta(\hat{H} - \mu \hat{N})} \tag{4.7}$$

thus it follows

$$T dS = \sum_i E_i dp_i - \mu \sum_i N_i dp_i \tag{4.8}$$

then notice that

$$d\langle N\rangle = \sum_i N_i dp_i - \mu \sum_i p_i dN_i = \sum_i N_i dp_i \tag{4.9}$$

where we have used $dN_i = 0$, as it does not depend on external parameters.
Thus we have

$$dU = TdS - PdV + \mu dn \tag{4.10}$$

where we have also used $n = \langle N\rangle / N_A$, and multipled our original definition of chemical potential $\mu N_A \to \mu$ the equation can be generalised to multiple species

$$dU = TdS - PdV + \sum_J \mu_J dn_J \tag{4.11}$$

entropy is given by an expression analogous to Eq. (3.12)

$$S = k_B \frac{\partial}{\partial T}(T \ln \mathcal{Z}) \tag{4.12}$$

and

$$U - \mu\langle N\rangle = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} \tag{4.13}$$

$$\langle N\rangle = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial \mu} \tag{4.14}$$

$$\sigma_N^2 = \frac{1}{\beta^2}\frac{\partial^2 \ln \mathcal{Z}}{\partial \mu^2} = \frac{1}{\beta}\frac{\partial \langle N\rangle}{\partial \mu} \tag{4.15}$$

where the fluctuation of the number of particles is related to variation of the mean particle number with respect to chemical potential, this is analogous to the earlier case where the fluctuation in energy is related to heat capacity.

Perhaps the most experimentally accessible ensamble is constant $N, p, T$, the **isobaric-isothermal emsamble**, where we now allow the volume to fluctuate. However, this leads to the question of what is a volume operator in quantum mechanics, as far as we are concerned, it does not exist. (Actually, introducing a volume operator would lead to quantum gravity, but that is a bit beyond the scope of these notes) We write the partition function as

$$\Delta = \int_0^\infty dV\, e^{-\beta PV} \operatorname{tr}\left[e^{-\beta \hat{H}(V)}\right] \tag{4.16}$$

$$p(i|V) = \frac{1}{Q(V)} e^{-\beta E_i(V)} \tag{4.17}$$

which the probability of the system being in state $i$ given that the system has volume $V$, and the probability of the sytem having volume $V \to V + dV$ is

$$p(V)dV = \frac{1}{\Delta} Q(V) e^{-\beta PV} dV \tag{4.18}$$

thus the joint probability is

$$p(i|V)p(V)dV = \frac{1}{\Delta}e^{-\beta(E_i(V)+PV)}dV \tag{4.19}$$

where $p(V)$ is a continuous function of $V$ and $P(i|V)$ is descrete.

the free energy with the closest relation to the partition function is the **Gibbs free energy**

$$G = -k_B T \ln \Delta \tag{4.20}$$

and it also follows

$$-\frac{\partial \ln \Delta}{\partial \beta} = \sum_i p_i(E_i + PV_i) = U + P\langle V \rangle \tag{4.21}$$

we define **Enthalpy** as

$$H = U + P\langle V \rangle = \langle E + PV \rangle \tag{4.22}$$

thus we have

$$H = -\frac{\partial \ln \Delta}{\partial \beta} \tag{4.23}$$

and similar to before

$$\sigma_H^2 = \frac{\partial^2 \ln \Delta}{\partial \beta^2} = k_B T^2 C_P \tag{4.24}$$

where we have defined isobaric heat capacity as

$$C_P = \Big(\frac{\partial H}{\partial T}\Big)_P \tag{4.25}$$

which relates to the fluctuation in enthalpy.
which can also be written as

$$dG = VdP - SdT + \sum_J \mu_J dn_J \tag{4.26}$$

where **Gibbs free energy** is defined as

$$G = A + PV \tag{4.27}$$

which is also

$$G = \sum_J \mu_J n_J \tag{4.28}$$

and it follows that

$$\Phi = A - G = -PV \tag{4.29}$$

thus chemical potential can be written as

$$\mu_J = \Big(\frac{\partial G}{\partial n_J}\Big)_{p,T,n_{J'}} \tag{4.30}$$

where $J'$ means all species other than $J$, which coincides with the definition in classical thermodynamics.